

## Multiple Choice

1. **Answer:** B

*Options:* A) Prevalence rates are 2% for female adolescents; B) Males from any socioeconomic background can be affected; C) Prevalence rates are 10% for male adolescents; D) It only affects individuals from high socioeconomic backgrounds; E) It affects males and females equally

*Note:* The correct answer is supported by the fact that Anorexia Nervosa can affect individuals regardless of gender or socioeconomic status, emphasizing its prevalence across diverse populations, including males from various backgrounds.

2. **Answer:** C

*Options:* A) Hemoglobin levels.; B) Central venous pressure.; C) Temperature.; D) Bicarbonate levels.; E) Urine output.

*Note:* Temperature is a critical physiological parameter that reflects the body's response to various neurological conditions and is routinely monitored to assess a patient's health status in a clinical setting.

3. **Answer:** A

*Options:* A) Values; B) Communication skills; C) Identity; D) Traits; E) Self-esteem

*Note:* Values tend to evolve significantly over time due to life experiences, cultural shifts, and personal development, reflecting the dynamic nature of individual priorities as people age.

4. **Answer:** A

*Options:* A) Vitamin D supplements; B) Fruit and vegetables; C) Salt; D) Red meat; E) Dairy products

*Note:* Research has indicated that high levels of vitamin D may influence cellular growth and differentiation, potentially impacting the development of certain cancers, including esophageal cancer, thus suggesting a correlation between vitamin D supplements and an increased risk for this type of cancer.

5. **Answer:** A

*Options:* A) No change in the first two trimesters, then an additional 500 kCal in the third trimester; B) No change from pre-pregnant levels; C) A decrease of 200 kCal throughout pregnancy; D) An additional 300 kCal in the second and third trimesters only; E) Less than 200 kCal

*Note:* The correct answer reflects the established nutritional guidelines that indicate a woman's caloric needs do not significantly increase during the first two trimesters of pregnancy, but require an increase of approximately 500 calories per day in the third trimester to support fetal growth and development.

## True / False

6. **Answer:** True

*Options:* A) True; B) False

*Note:* Parasympathetic preganglionic nerves primarily exit the central nervous system via cranial nerves III (oculomotor), VII (facial), IX (glossopharyngeal), and X (vagus), with the fourth cranial nerve (trochlear) not being involved, but the statement may refer to the general distribution of parasympathetic fibers rather than specific cranial nerves, hence the potential confusion in classification.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* Fructose is absorbed in the gastrointestinal tract primarily through the GLUT 5 transporter, whereas glucose and galactose are absorbed via the sodium-glucose co-transporter (SGLT 1).

8. **Answer:** False

*Options:* A) True; B) False

*Note:* Athletic ability typically declines in adulthood due to physiological changes such as decreased muscle mass, strength, and cardiovascular function, as well as decreased physical activity levels.

9. **Answer:** True

*Options:* A) True; B) False

*Note:* Muscle glycogen is converted into glucose and subsequently metabolized to provide the rapid energy required for ATP production during high-intensity, short-duration activities such as sprinting.

10. **Answer:** False

*Options:* A) True; B) False

*Note:* The statement is false because while some natural polyunsaturated fatty acids do contain 20 carbon atoms, they typically have multiple double bonds (such as in arachidonic acid), but not limited to just two double bonds, and the more common 20-carbon polyunsaturated fatty acids, like eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), have five and six double bonds, respectively.

## Long Form Response

11. **Answer:** The gastrointestinal tract plays a critical role in survival by facilitating the elimination of toxins immediately after a meal, thereby protecting the body from potential harm. Upon ingestion, the digestive system not only breaks down food but also processes and neutralizes harmful substances, such as bacteria, chemicals, and metabolic byproducts. The physiological significance of this function lies in the gut's ability to maintain homeostasis, prevent systemic absorption of toxins, and support immune responses through the action of gut-associated lymphoid tissue (GALT). Efficient elimination through mechanisms such as peristalsis and the secretion of bile and

digestive enzymes ensures that any ingested toxins are swiftly excreted, minimizing their impact on overall health. This detoxification process is crucial for maintaining metabolic balance and safeguarding the integrity of bodily functions, highlighting the importance of the gastrointestinal tract in health and survival.

*Note:* correct because it emphasizes the gastrointestinal tract's essential functions in detoxifying harmful substances and maintaining homeostasis immediately after food intake, which is vital for the body's overall survival and health.

12. **Answer:** Male-to-male transmission plays a significant role in the context of autosomal incomplete dominance, as it highlights the inheritance patterns of traits that do not follow traditional Mendelian ratios. In autosomal incomplete dominance, the phenotype of heterozygotes is intermediate between those of the homozygotes, which can lead to varying expression of genetic traits among male offspring. This is particularly important in genetic counseling, as understanding these patterns can help predict the likelihood of disease prevalence in male descendants, especially for conditions influenced by incomplete dominance. Consequently, genetic counselors must consider these complexities when advising families, as they influence the genetic risk assessment and management strategies for conditions that may exhibit sex-biased inheritance. Ultimately, recognizing the implications of male-to-male transmission in autosomal incomplete dominance allows for more tailored and informed genetic counseling practices.

*Note:* correct because it emphasizes how male-to-male transmission in autosomal incomplete dominance affects phenotype expression and inheritance patterns, which are crucial for accurate genetic counseling and understanding disease prevalence.